The Jelly Ocean Hypothesis in the Northwest North Atlantic

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Introduction

In the study of zooplankton ecology, there has often been a "crustaceocentric" bias (Haddock 2004). The study of gelatinous species, collectively termed gelata, has been underemphasized as compared to crustaceans. In the early 2000s, gelata moved into the spotlight, following conspicuous blooms that damaged fisheries (Kideys et al. 2005), aquaculture (Clinton et al. 2021), energy production (Wang et al. 2023), and recreation (Nunes et al. 2015). In addition to motivating new research on gelata, these events raised the question of whether there was a larger pattern of change emerging.

Emerging from a series of studies was the Jelly Ocean Hypothesis, which posits that gelatinous zooplankton populations are increasing worldwide as a result of changing ocean conditions, primarily due to anthropogenic activities. These studies argued that due a multiplicity of causes, abundance of gelatinous species is growing. Proposed causes are primarily human-driven, including warming (ref), eutrophication, translocation (Kideys et al. 2005), overfishing, bottom trawling (Qui 2014), pollution, aquaculture, and acidification (Richardson and Gibbons 2008). There is also evidence that shifts toward dominance of gelata are difficult to reverse because of dynamics like histeresis (Richardson et al. 2009). The hypothesis echos a concern that has been around for centuries; the 19th century novelist Jules Verne once predicted that, "crowded with jellyfish, squid, and other devilfish, the oceans will have become huge centers of infection" (REF).

Testing the Jelly Ocean Hypothesis requires long time series (>20 years) of gelata abundances in order to compare to historical levels. The shortage of such datasets has left the Jelly Ocean Hypothesis largely unsupported (Condon et al. 2012). In lieu of time series, studies have used qualitative data such as news reports to argue that gelata are increasing globally (Brotz et al. 2012). However, it is difficult to distinguish these perceived increases from multi-annual or multi-decadal fluctuations. A recent meta-analysis of time series showed that most perceived increases appear to reflect a ~20 year cyclicity, with a worldwide upswing in the 1990s, coinciding with the increase in concern over gelata (Condon et al. 2013), although some notable long-term increases in gelata were not included in the analysis (Atkinson et al. 2004). A metaanalysis of corpus of research on this topic found flawed citations and misinterpreted conclusions (Sanz-Martin et al. 2016), and at this point, the Jelly Ocean Hypothesis remains very much an open question.

In some ways, the Jelly Ocean Hypothesis is loosely posed. Gelata are taxonomically very diverse, encompassing multiple phyla. This diversity confounds attempts to quantify increasing trends. The list of species classified under the gelatinous umbrella varies, and includes taxa with drastic and important differences in their life histories, trophic positions, and ecological roles. For example, pelagic ctenophores are primarily copepod predators, where as pelagic tunicates graze some of the smallest microbes. Cnidarians and ctenophores are the two most common taxa to group together in analysis, and yet they are phylogenetically very distant (Moroz et al. 2014). Perhaps the only characteristic common to the many gelatinous phyla is a body with a high water content, alternatively expressed as a low carbon:volume ratio. There is no a priori reason to expect that phyla with such striking differences and perhaps only a superficial similarity should be grouped together with respect to their temporal trends. Some have suggested that the Jelly Ocean Hypothesis is ill-posed—i.e. which species are increasing, where, and for what reason (Gibbons & Richardson 2013)?

Despite their major differences, there is some evidence that a diverse collection of gelata can vary together (Condon et al. 2013). This suggests that certain conditions can occur that favor the gelatinous body form in general, over other strategies that utilize a high carbon:volume ratio (e.g. crustaceans). This reframes the Jelly Ocean Hypothesis in trait-based terms: conditions in the ocean ecosystem can shift to benefit the gelatinous body form. This would be inclusive of gelata in general, despite differences in phylogeny, trophic position, or other life history traits.

We revisited the Jelly Ocean Hypothesis using a 50-year spatial survey on the northwest North Atlantic Shelf. We addressed two main questions: 1) what are the temporal changes in gelata presence; and 2) what are the underlying conditions associated with changes. We examined three taxonomic groups well-represented in the dataset–siphonophores, salps, and coelenterates–to explore the idea that certain conditions can favor the gelatinous body plan in general.

Methods

Data

The Northeast Fisheries Science Center Ecosystem Monitoring (ECOMON) program collects seasonal zooplankton measurements as part of its mission to monitor and assess the Northeast Continental Shelf ecosystem. Sampling includes oblique tows using a 61 cm bongo net fitted with a 333 µm mesh at stratified locations across the shelf (Figure 1). Tows are a minimum of five minutes in duration and sample from the surface to within 5 m of the seabed or to a maximum depth of 200 m. A mechanical flowmeter fitted in the mouth of each net is used to quantify volume sampled. Gelatinous zooplankton are identified into broad taxonomic groups, including siphonophores, ctenophores, salps, and coelenterates (i.e. scyphozoans). The broad grouping, while missing … has allowed for internal consistency over the duration of the survey. The analysis conducted here uses presence or absence of each taxonomic group.

Time series analysis

Spatial modeling

Results

Discussion

References

Atkinson, Angus, Volker Siegel, Evgeny Pakhomov, and Peter Rothery. 2004. "Long-Term Decline in Krill Stock and Increase in Salps Within the Southern Ocean." *Nature* 432 (7013): 100–103.

Brotz, Lucas, William WL Cheung, Kristin Kleisner, Evgeny Pakhomov, and Daniel Pauly. 2012. "Increasing Jellyfish Populations: Trends in Large Marine Ecosystems." In *Jellyfish Blooms IV: Interactions with Humans and Fisheries*, 3–20. Springer.

Clinton, Morag, David EK Ferrier, Samuel AM Martin, and Andrew S Brierley. 2021. "Impacts of Jellyfish on Marine Cage Aquaculture: An Overview of Existing Knowledge and the Challenges to Finfish Health." *ICES Journal of Marine Science* 78 (5): 1557–73.

Condon, Robert H, Carlos M Duarte, Kylie A Pitt, Kelly L Robinson, Cathy H Lucas, Kelly R Sutherland, Hermes W Mianzan, et al. 2013. "Recurrent Jellyfish Blooms Are a Consequence of Global Oscillations." *Proceedings of the National Academy of Sciences* 110 (3): 1000–1005.

Condon, Robert H, William M Graham, Carlos M Duarte, Kylie A Pitt, Cathy H Lucas, Steven HD Haddock, Kelly R Sutherland, et al. 2012. "Questioning the Rise of Gelatinous Zooplankton in the World's Oceans." *Bioscience* 62 (2): 160–69.

Haddock, Steven HD. 2004. "A Golden Age of Gelata: Past and Future Research on Planktonic Ctenophores and Cnidarians." *Hydrobiologia* 530 (1): 549–56.

Kideys, Ahmet Erkan, Abolchaseem Roohi, Siamak Bagheri, Galina Finenko, and Lyudmila Kamburska. 2005. "Impacts of Invasive Ctenophores on the Fisheries of the Black Sea and Caspian Sea." *Oceanography* 18 (2): 76–85.

Nunes, Paulo ALD, Maria L Loureiro, Laia Piñol, Sergio Sastre, Louinord Voltaire, and Antonio Canepa. 2015. "Analyzing Beach Recreationists' Preferences for the Reduction of Jellyfish Blooms: Economic Results from a Stated-Choice Experiment in Catalonia, Spain." *PloS One* 10 (6): e0126681.

Richardson, Anthony J, Andrew Bakun, Graeme C Hays, and Mark J Gibbons. 2009. "The Jellyfish Joyride: Causes, Consequences and Management Responses to a More Gelatinous Future." *Trends in Ecology & Evolution* 24 (6): 312–22.

Richardson, Anthony J, and Mark J Gibbons. 2008. "Are Jellyfish Increasing in Response to Ocean Acidification?" *Limnology and Oceanography* 53 (5): 2040–45.

Sanz-Martin, Marina, Kylie A Pitt, Robert H Condon, Cathy H Lucas, Charles Novaes de Santana, and Carlos M Duarte. 2016. "Flawed Citation Practices Facilitate the Unsubstantiated Perception of a Global Trend Toward Increased Jellyfish Blooms." *Global Ecology and Biogeography* 25 (9): 1039–49.

Wang, Xiaocheng, Qingqing Jin, Lu Yang, Chuan Jia, Chunjiang Guan, Haining Wang, and Hao Guo. 2023. "Aggregation Process of Two Disaster-Causing Jellyfish Species, Nemopilema Nomurai and Aurelia Coerulea, at the Intake Area of a Nuclear Power Cooling-Water System in Eastern Liaodong Bay, China." *Frontiers in Marine Science* 9: 1098232.